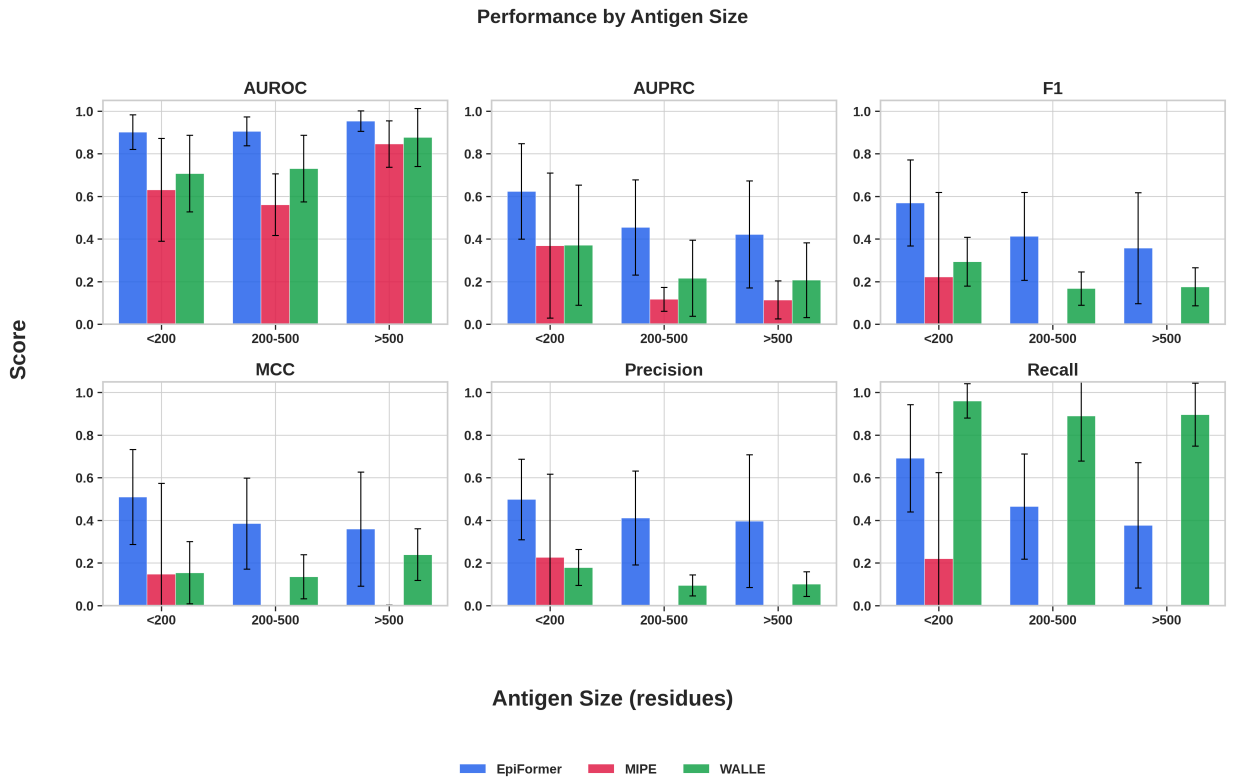

Figure 1. Per-complex AUC and F1 box plot distributions stratified by antigen surface residue count ($<$ 200, 200–500, $>$ 500) for EpiFormer, WALLE, and MIPE on the AsEP epitope_ratio test set. EpiFormer's ranking ability (AUC) improves with antigen size, while F1 declines due to increasing class imbalance (mean epitope ratio drops from 13.9% to 2.2%). EpiFormer maintains its advantage over baselines at all size strata. MIPE collapses to F1=0 for medium and large antigens.

Table 1. Performance stratified by antigen surface residue count on the AsEP epitope_ratio test set (per-complex mean $\pm$ std). Complexes are binned into small ($<$ 200 residues, 90 complexes), medium (200–500, 62), and large ($>$ 500, 18). Mean epitope ratio decreases with antigen size (13.9% $\rightarrow$ 6.6% $\rightarrow$ 2.2%), making larger antigens inherently harder due to extreme class imbalance.

Size Bin	N	Epi. Ratio	Method	AUC	F1	MCC
			EpiFormer	**.901±.081**	**.568±.202**	**.509±.222**
$<$ 200 res. (small)	90	13.9%	WALLE	.707±.180	.293±.114	.154±.146
			MIPE	.630±.241	.222±.396	.147±.425
			EpiFormer	**.905±.068**	**.412±.206**	**.384±.214**
200–500 res. (med)	62	6.6%	WALLE	.730±.156	.167±.078	.135±.103
			MIPE	.561±.145	.000±.000	.000±.000
			EpiFormer	**.953±.047**	**.356±.260**	**.358±.268**
$>$ 500 res. (large)	18	2.2%	WALLE	.876±.136	.175±.090	.239±.121
			MIPE	.845±.109	.000±.000	.003 ± .005

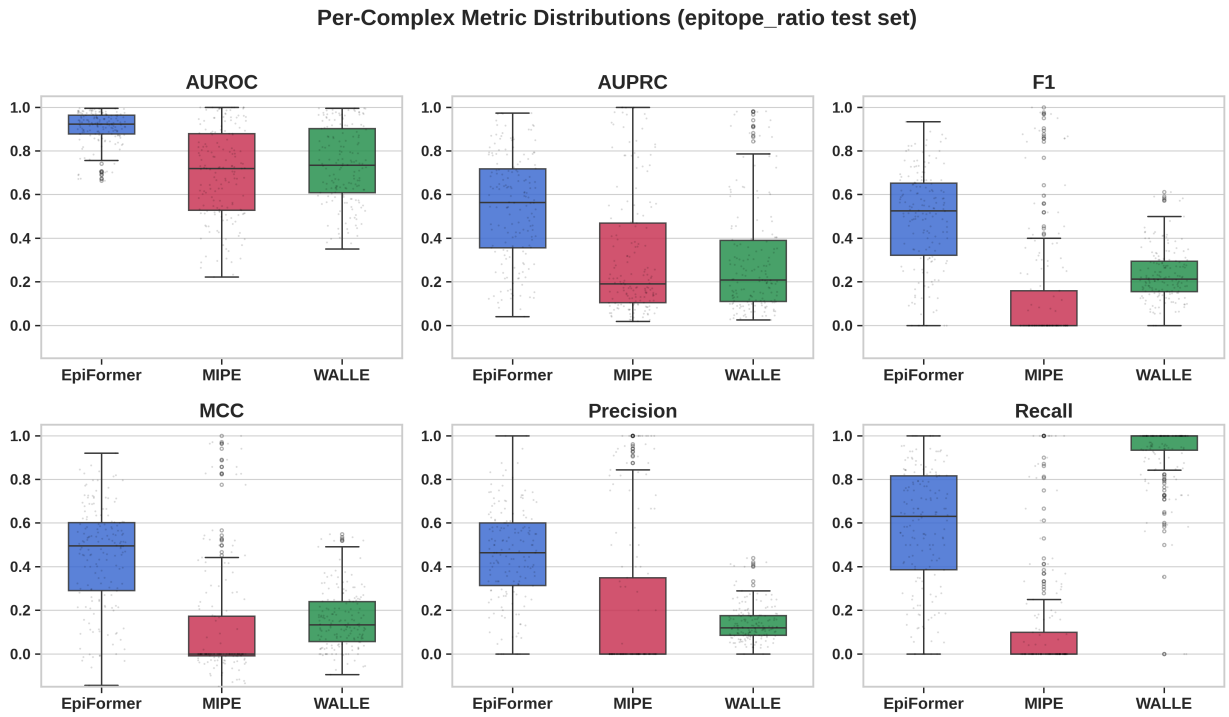

Figure 2. Per-complex metric distributions (AUC, AUPRC, F1, MCC, Precision, Recall) across the full AsEP epitope_ratio test set (170 complexes) for EpiFormer, MIPE, and WALLE. Each box shows median, interquartile range, and outliers. EpiFormer achieves higher median and tighter variance across all 6 metrics compared to both baselines.